# SG No. 219 $T_{\mathrm{d}}^{5}$ $F\bar{4}3c$ [ cubic ]

* plus set: $+[0,\,0,\,0]$, $+\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$, $+\left[\frac{1}{2},\,\frac{1}{2},\,0\right]$
* Wyckoff site: `8a`, site symmetry: `23.`

Table 1: Wyckoff bond: `24a@8a`, bond symmetry: `222..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X,\,0,\,0]$ | $[0,\,0,\,0]$ | `[1,-2,-3,4]` |
| 2 | $[0,\,X,\,0]$ | $[0,\,0,\,0]$ | `[5,-6,-7,8]` |
| 3 | $[0,\,0,\,X]$ | $[0,\,0,\,0]$ | `[9,-10,-11,12]` |
| 4 | $[0,\,X,\,0]$ | $\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$ | `[13,-14,-15,16]` |
| 5 | $[X,\,0,\,0]$ | $\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$ | `[17,-18,-19,20]` |
| 6 | $[0,\,0,\,X]$ | $\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$ | `[21,-22,-23,24]` |

Table 2: Wyckoff bond: `32b@8a`, bond symmetry: `.3.`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X,\,X,\,X]$ | $[0,\,0,\,0]$ | `[1,5,9]` |
| 2 | $[-X,\,-X,\,X]$ | $[0,\,0,\,0]$ | `[2,7,12]` |
| 3 | $[-X,\,X,\,-X]$ | $[0,\,0,\,0]$ | `[3,8,10]` |
| 4 | $[X,\,-X,\,-X]$ | $[0,\,0,\,0]$ | `[4,6,11]` |
| 5 | $[X,\,X,\,X]$ | $\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$ | `[13,17,21]` |
| 6 | $[-X,\,-X,\,X]$ | $\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$ | `[14,19,24]` |
| 7 | $[X,\,-X,\,-X]$ | $\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$ | `[15,20,22]` |
| 8 | $[-X,\,X,\,-X]$ | $\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$ | `[16,18,23]` |

Table 3: Wyckoff bond: `48c@8a`, bond symmetry: `2..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X,\,Y,\,0]$ | $[0,\,0,\,0]$ | `[1,-2]` |
| 2 | $[-X,\,Y,\,0]$ | $[0,\,0,\,0]$ | `[3,-4]` |
| 3 | $[0,\,X,\,Y]$ | $[0,\,0,\,0]$ | `[5,-6]` |
| 4 | $[0,\,-X,\,Y]$ | $[0,\,0,\,0]$ | `[7,-8]` |
| 5 | $[Y,\,0,\,X]$ | $[0,\,0,\,0]$ | `[9,-10]` |
| 6 | $[Y,\,0,\,-X]$ | $[0,\,0,\,0]$ | `[11,-12]` |
| 7 | $[Y,\,X,\,0]$ | $\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$ | `[13,-14]` |
| 8 | $[Y,\,-X,\,0]$ | $\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$ | `[15,-16]` |
| 9 | $[X,\,0,\,Y]$ | $\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$ | `[17,-18]` |
| 10 | $[-X,\,0,\,Y]$ | $\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$ | `[19,-20]` |
| 11 | $[0,\,Y,\,X]$ | $\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$ | `[21,-22]` |
| 12 | $[0,\,Y,\,-X]$ | $\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$ | `[23,-24]` |

Table 4: Wyckoff bond: `96d@8a`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X,\,Y,\,Z]$ | $[0,\,0,\,0]$ | `[1]` |

*continued ...*

Table 4

| No. | vector | center | mapping |
|---|---|---|---|
| 2 | $[-X, -Y, Z]$ | [0, 0, 0] | [2] |
| 3 | $[-X, Y, -Z]$ | [0, 0, 0] | [3] |
| 4 | $[X, -Y, -Z]$ | [0, 0, 0] | [4] |
| 5 | $[Z, X, Y]$ | [0, 0, 0] | [5] |
| 6 | $[Z, -X, -Y]$ | [0, 0, 0] | [6] |
| 7 | $[-Z, -X, Y]$ | [0, 0, 0] | [7] |
| 8 | $[-Z, X, -Y]$ | [0, 0, 0] | [8] |
| 9 | $[Y, Z, X]$ | [0, 0, 0] | [9] |
| 10 | $[-Y, Z, -X]$ | [0, 0, 0] | [10] |
| 11 | $[Y, -Z, -X]$ | [0, 0, 0] | [11] |
| 12 | $[-Y, -Z, X]$ | [0, 0, 0] | [12] |
| 13 | $[Y, X, Z]$ | $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$ | [13] |
| 14 | $[-Y, -X, Z]$ | $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$ | [14] |
| 15 | $[Y, -X, -Z]$ | $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$ | [15] |
| 16 | $[-Y, X, -Z]$ | $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$ | [16] |
| 17 | $[X, Z, Y]$ | $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$ | [17] |
| 18 | $[-X, Z, -Y]$ | $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$ | [18] |
| 19 | $[-X, -Z, Y]$ | $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$ | [19] |
| 20 | $[X, -Z, -Y]$ | $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$ | [20] |
| 21 | $[Z, Y, X]$ | $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$ | [21] |
| 22 | $[Z, -Y, -X]$ | $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$ | [22] |
| 23 | $[-Z, Y, -X]$ | $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$ | [23] |
| 24 | $[-Z, -Y, X]$ | $[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$ | [24] |

* Wyckoff site: 8b, site symmetry: 23.

Table 5: Wyckoff bond: 24a@8b, bond symmetry: 222..

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, 0, 0]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | [1,-2,-3,4] |
| 2 | $[0, X, 0]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | [5,-6,-7,8] |
| 3 | $[0, 0, X]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | [9,-10,-11,12] |
| 4 | $[0, X, 0]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | [13,-14,-15,16] |
| 5 | $[X, 0, 0]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | [17,-18,-19,20] |
| 6 | $[0, 0, X]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | [21,-22,-23,24] |

Table 6: Wyckoff bond: 32b@8b, bond symmetry: .3.

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, X, X]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | [1,5,9] |
| 2 | $[-X, -X, X]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | [2,7,12] |
| 3 | $[-X, X, -X]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | [3,8,10] |
| 4 | $[X, -X, -X]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | [4,6,11] |
| 5 | $[X, X, X]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | [13,17,21] |
| 6 | $[-X, -X, X]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | [14,19,24] |
| 7 | $[X, -X, -X]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | [15,20,22] |

*continued ...*

Table 6

| No. | vector | center | mapping |
|---|---|---|---|
| 8 | $[-X, X, -X]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[16,18,23]` |

Table 7: Wyckoff bond: `48c@8b`, bond symmetry: `2..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, 0]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[1,-2]` |
| 2 | $[-X, Y, 0]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[3,-4]` |
| 3 | $[0, X, Y]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[5,-6]` |
| 4 | $[0, -X, Y]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[7,-8]` |
| 5 | $[Y, 0, X]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[9,-10]` |
| 6 | $[Y, 0, -X]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[11,-12]` |
| 7 | $[Y, X, 0]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[13,-14]` |
| 8 | $[Y, -X, 0]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[15,-16]` |
| 9 | $[X, 0, Y]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[17,-18]` |
| 10 | $[-X, 0, Y]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[19,-20]` |
| 11 | $[0, Y, X]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[21,-22]` |
| 12 | $[0, Y, -X]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[23,-24]` |

Table 8: Wyckoff bond: `96d@8b`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[2]` |
| 3 | $[-X, Y, -Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[3]` |
| 4 | $[X, -Y, -Z]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[4]` |
| 5 | $[Z, X, Y]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[5]` |
| 6 | $[Z, -X, -Y]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[6]` |
| 7 | $[-Z, -X, Y]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[7]` |
| 8 | $[-Z, X, -Y]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[8]` |
| 9 | $[Y, Z, X]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[9]` |
| 10 | $[-Y, Z, -X]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[10]` |
| 11 | $[Y, -Z, -X]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[11]` |
| 12 | $[-Y, -Z, X]$ | $[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$ | `[12]` |
| 13 | $[Y, X, Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[13]` |
| 14 | $[-Y, -X, Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[14]` |
| 15 | $[Y, -X, -Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[15]` |
| 16 | $[-Y, X, -Z]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[16]` |
| 17 | $[X, Z, Y]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[17]` |
| 18 | $[-X, Z, -Y]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[18]` |
| 19 | $[-X, -Z, Y]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[19]` |
| 20 | $[X, -Z, -Y]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[20]` |
| 21 | $[Z, Y, X]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[21]` |
| 22 | $[Z, -Y, -X]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[22]` |
| 23 | $[-Z, Y, -X]$ | $[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$ | `[23]` |

*continued ...*

Table 8

| No. | vector | center | mapping |
|---|---|---|---|
| 24 | $[-Z, -Y, X]$ | $\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | `[24]` |

* Wyckoff site: `24c`, site symmetry: `-4..`

Table 9: Wyckoff bond: `24a@24c`, bond symmetry: `-4..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, 0, 0]$ | $\left[0, \frac{1}{4}, \frac{1}{4}\right]$ | `[1,4,-18,-19]` |
| 2 | $[-X, 0, 0]$ | $\left[0, \frac{3}{4}, \frac{1}{4}\right]$ | `[2,3,-17,-20]` |
| 3 | $[0, X, 0]$ | $\left[\frac{1}{4}, 0, \frac{1}{4}\right]$ | `[5,8,-14,-15]` |
| 4 | $[0, -X, 0]$ | $\left[\frac{1}{4}, 0, \frac{3}{4}\right]$ | `[6,7,-13,-16]` |
| 5 | $[0, 0, X]$ | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[9,12,-22,-23]` |
| 6 | $[0, 0, -X]$ | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[10,11,-21,-24]` |

Table 10: Wyckoff bond: `48b@24c`, bond symmetry: `2..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, X, Y]$ | $\left[0, \frac{1}{4}, \frac{1}{4}\right]$ | `[1,-4]` |
| 2 | $[0, -X, Y]$ | $\left[0, \frac{3}{4}, \frac{1}{4}\right]$ | `[2,-3]` |
| 3 | $[Y, 0, X]$ | $\left[\frac{1}{4}, 0, \frac{1}{4}\right]$ | `[5,-8]` |
| 4 | $[Y, 0, -X]$ | $\left[\frac{1}{4}, 0, \frac{3}{4}\right]$ | `[6,-7]` |
| 5 | $[X, Y, 0]$ | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[9,-12]` |
| 6 | $[-X, Y, 0]$ | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[10,-11]` |
| 7 | $[X, 0, Y]$ | $\left[\frac{1}{4}, 0, \frac{3}{4}\right]$ | `[13,-16]` |
| 8 | $[-X, 0, Y]$ | $\left[\frac{1}{4}, 0, \frac{1}{4}\right]$ | `[14,-15]` |
| 9 | $[0, Y, X]$ | $\left[0, \frac{3}{4}, \frac{1}{4}\right]$ | `[17,-20]` |
| 10 | $[0, Y, -X]$ | $\left[0, \frac{1}{4}, \frac{1}{4}\right]$ | `[18,-19]` |
| 11 | $[Y, X, 0]$ | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[21,-24]` |
| 12 | $[Y, -X, 0]$ | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[22,-23]` |

Table 11: Wyckoff bond: `96c@24c`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $\left[0, \frac{1}{4}, \frac{1}{4}\right]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $\left[0, \frac{3}{4}, \frac{1}{4}\right]$ | `[2]` |
| 3 | $[-X, Y, -Z]$ | $\left[0, \frac{3}{4}, \frac{1}{4}\right]$ | `[3]` |
| 4 | $[X, -Y, -Z]$ | $\left[0, \frac{1}{4}, \frac{1}{4}\right]$ | `[4]` |
| 5 | $[Z, X, Y]$ | $\left[\frac{1}{4}, 0, \frac{1}{4}\right]$ | `[5]` |
| 6 | $[Z, -X, -Y]$ | $\left[\frac{1}{4}, 0, \frac{3}{4}\right]$ | `[6]` |
| 7 | $[-Z, -X, Y]$ | $\left[\frac{1}{4}, 0, \frac{3}{4}\right]$ | `[7]` |
| 8 | $[-Z, X, -Y]$ | $\left[\frac{1}{4}, 0, \frac{1}{4}\right]$ | `[8]` |
| 9 | $[Y, Z, X]$ | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[9]` |
| 10 | $[-Y, Z, -X]$ | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[10]` |
| 11 | $[Y, -Z, -X]$ | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[11]` |

*continued ...*

Table 11

| No. | vector | center | mapping |
|---|---|---|---|
| 12 | $[-Y, -Z, X]$ | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[12]` |
| 13 | $[Y, X, Z]$ | $\left[\frac{1}{4}, 0, \frac{3}{4}\right]$ | `[13]` |
| 14 | $[-Y, -X, Z]$ | $\left[\frac{1}{4}, 0, \frac{1}{4}\right]$ | `[14]` |
| 15 | $[Y, -X, -Z]$ | $\left[\frac{1}{4}, 0, \frac{1}{4}\right]$ | `[15]` |
| 16 | $[-Y, X, -Z]$ | $\left[\frac{1}{4}, 0, \frac{3}{4}\right]$ | `[16]` |
| 17 | $[X, Z, Y]$ | $\left[0, \frac{3}{4}, \frac{1}{4}\right]$ | `[17]` |
| 18 | $[-X, Z, -Y]$ | $\left[0, \frac{1}{4}, \frac{1}{4}\right]$ | `[18]` |
| 19 | $[-X, -Z, Y]$ | $\left[0, \frac{1}{4}, \frac{1}{4}\right]$ | `[19]` |
| 20 | $[X, -Z, -Y]$ | $\left[0, \frac{3}{4}, \frac{1}{4}\right]$ | `[20]` |
| 21 | $[Z, Y, X]$ | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[21]` |
| 22 | $[Z, -Y, -X]$ | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[22]` |
| 23 | $[-Z, Y, -X]$ | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | `[23]` |
| 24 | $[-Z, -Y, X]$ | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[24]` |

* Wyckoff site: `24d`, site symmetry: `-4..`

Table 12: Wyckoff bond: `24a@24d`, bond symmetry: `-4..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, 0, 0]$ | $\left[\frac{1}{4}, 0, 0\right]$ | `[1,4,-18,-19]` |
| 2 | $[-X, 0, 0]$ | $\left[\frac{3}{4}, 0, 0\right]$ | `[2,3,-17,-20]` |
| 3 | $[0, X, 0]$ | $\left[0, \frac{1}{4}, 0\right]$ | `[5,8,-14,-15]` |
| 4 | $[0, -X, 0]$ | $\left[0, \frac{3}{4}, 0\right]$ | `[6,7,-13,-16]` |
| 5 | $[0, 0, X]$ | $\left[0, 0, \frac{1}{4}\right]$ | `[9,12,-22,-23]` |
| 6 | $[0, 0, -X]$ | $\left[0, 0, \frac{3}{4}\right]$ | `[10,11,-21,-24]` |

Table 13: Wyckoff bond: `48b@24d`, bond symmetry: `2..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, X, Y]$ | $\left[\frac{1}{4}, 0, 0\right]$ | `[1,-4]` |
| 2 | $[0, -X, Y]$ | $\left[\frac{3}{4}, 0, 0\right]$ | `[2,-3]` |
| 3 | $[Y, 0, X]$ | $\left[0, \frac{1}{4}, 0\right]$ | `[5,-8]` |
| 4 | $[Y, 0, -X]$ | $\left[0, \frac{3}{4}, 0\right]$ | `[6,-7]` |
| 5 | $[X, Y, 0]$ | $\left[0, 0, \frac{1}{4}\right]$ | `[9,-12]` |
| 6 | $[-X, Y, 0]$ | $\left[0, 0, \frac{3}{4}\right]$ | `[10,-11]` |
| 7 | $[X, 0, Y]$ | $\left[0, \frac{3}{4}, 0\right]$ | `[13,-16]` |
| 8 | $[-X, 0, Y]$ | $\left[0, \frac{1}{4}, 0\right]$ | `[14,-15]` |
| 9 | $[0, Y, X]$ | $\left[\frac{3}{4}, 0, 0\right]$ | `[17,-20]` |
| 10 | $[0, Y, -X]$ | $\left[\frac{1}{4}, 0, 0\right]$ | `[18,-19]` |
| 11 | $[Y, X, 0]$ | $\left[0, 0, \frac{3}{4}\right]$ | `[21,-24]` |
| 12 | $[Y, -X, 0]$ | $\left[0, 0, \frac{1}{4}\right]$ | `[22,-23]` |

Table 14: Wyckoff bond: `96c@24d`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $\left[\frac{1}{4}, 0, 0\right]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $\left[\frac{3}{4}, 0, 0\right]$ | `[2]` |
| 3 | $[-X, Y, -Z]$ | $\left[\frac{3}{4}, 0, 0\right]$ | `[3]` |
| 4 | $[X, -Y, -Z]$ | $\left[\frac{1}{4}, 0, 0\right]$ | `[4]` |
| 5 | $[Z, X, Y]$ | $\left[0, \frac{1}{4}, 0\right]$ | `[5]` |
| 6 | $[Z, -X, -Y]$ | $\left[0, \frac{3}{4}, 0\right]$ | `[6]` |
| 7 | $[-Z, -X, Y]$ | $\left[0, \frac{3}{4}, 0\right]$ | `[7]` |
| 8 | $[-Z, X, -Y]$ | $\left[0, \frac{1}{4}, 0\right]$ | `[8]` |
| 9 | $[Y, Z, X]$ | $\left[0, 0, \frac{1}{4}\right]$ | `[9]` |
| 10 | $[-Y, Z, -X]$ | $\left[0, 0, \frac{3}{4}\right]$ | `[10]` |
| 11 | $[Y, -Z, -X]$ | $\left[0, 0, \frac{3}{4}\right]$ | `[11]` |
| 12 | $[-Y, -Z, X]$ | $\left[0, 0, \frac{1}{4}\right]$ | `[12]` |
| 13 | $[Y, X, Z]$ | $\left[0, \frac{3}{4}, 0\right]$ | `[13]` |
| 14 | $[-Y, -X, Z]$ | $\left[0, \frac{1}{4}, 0\right]$ | `[14]` |
| 15 | $[Y, -X, -Z]$ | $\left[0, \frac{1}{4}, 0\right]$ | `[15]` |
| 16 | $[-Y, X, -Z]$ | $\left[0, \frac{3}{4}, 0\right]$ | `[16]` |
| 17 | $[X, Z, Y]$ | $\left[\frac{3}{4}, 0, 0\right]$ | `[17]` |
| 18 | $[-X, Z, -Y]$ | $\left[\frac{1}{4}, 0, 0\right]$ | `[18]` |
| 19 | $[-X, -Z, Y]$ | $\left[\frac{1}{4}, 0, 0\right]$ | `[19]` |
| 20 | $[X, -Z, -Y]$ | $\left[\frac{3}{4}, 0, 0\right]$ | `[20]` |
| 21 | $[Z, Y, X]$ | $\left[0, 0, \frac{3}{4}\right]$ | `[21]` |
| 22 | $[Z, -Y, -X]$ | $\left[0, 0, \frac{1}{4}\right]$ | `[22]` |
| 23 | $[-Z, Y, -X]$ | $\left[0, 0, \frac{1}{4}\right]$ | `[23]` |
| 24 | $[-Z, -Y, X]$ | $\left[0, 0, \frac{3}{4}\right]$ | `[24]` |

* Wyckoff site: `32e`, site symmetry: `.3.`

Table 15: Wyckoff bond: `32a@32e`, bond symmetry: `.3.`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, X, X]$ | $[x, x, x]$ | `[1,5,9]` |
| 2 | $[-X, -X, X]$ | $[-x, -x, x]$ | `[2,7,12]` |
| 3 | $[-X, X, -X]$ | $[-x, x, -x]$ | `[3,8,10]` |
| 4 | $[X, -X, -X]$ | $[x, -x, -x]$ | `[4,6,11]` |
| 5 | $[X, X, X]$ | $\left[x+\frac{1}{2}, x+\frac{1}{2}, x+\frac{1}{2}\right]$ | `[13,17,21]` |
| 6 | $[-X, -X, X]$ | $\left[\frac{1}{2}-x, \frac{1}{2}-x, x+\frac{1}{2}\right]$ | `[14,19,24]` |
| 7 | $[X, -X, -X]$ | $\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}-x\right]$ | `[15,20,22]` |
| 8 | $[-X, X, -X]$ | $\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}-x\right]$ | `[16,18,23]` |

Table 16: Wyckoff bond: `96b@32e`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $[x, x, x]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $[-x, -x, x]$ | `[2]` |
| 3 | $[-X, Y, -Z]$ | $[-x, x, -x]$ | `[3]` |

*continued ...*

Table 16

| No. | vector | center | mapping |
|---|---|---|---|
| 4 | $[X, -Y, -Z]$ | $[x, -x, -x]$ | [4] |
| 5 | $[Z, X, Y]$ | $[x, x, x]$ | [5] |
| 6 | $[Z, -X, -Y]$ | $[x, -x, -x]$ | [6] |
| 7 | $[-Z, -X, Y]$ | $[-x, -x, x]$ | [7] |
| 8 | $[-Z, X, -Y]$ | $[-x, x, -x]$ | [8] |
| 9 | $[Y, Z, X]$ | $[x, x, x]$ | [9] |
| 10 | $[-Y, Z, -X]$ | $[-x, x, -x]$ | [10] |
| 11 | $[Y, -Z, -X]$ | $[x, -x, -x]$ | [11] |
| 12 | $[-Y, -Z, X]$ | $[-x, -x, x]$ | [12] |
| 13 | $[Y, X, Z]$ | $\left[x+\frac{1}{2}, x+\frac{1}{2}, x+\frac{1}{2}\right]$ | [13] |
| 14 | $[-Y, -X, Z]$ | $\left[\frac{1}{2}-x, \frac{1}{2}-x, x+\frac{1}{2}\right]$ | [14] |
| 15 | $[Y, -X, -Z]$ | $\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}-x\right]$ | [15] |
| 16 | $[-Y, X, -Z]$ | $\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}-x\right]$ | [16] |
| 17 | $[X, Z, Y]$ | $\left[x+\frac{1}{2}, x+\frac{1}{2}, x+\frac{1}{2}\right]$ | [17] |
| 18 | $[-X, Z, -Y]$ | $\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}-x\right]$ | [18] |
| 19 | $[-X, -Z, Y]$ | $\left[\frac{1}{2}-x, \frac{1}{2}-x, x+\frac{1}{2}\right]$ | [19] |
| 20 | $[X, -Z, -Y]$ | $\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}-x\right]$ | [20] |
| 21 | $[Z, Y, X]$ | $\left[x+\frac{1}{2}, x+\frac{1}{2}, x+\frac{1}{2}\right]$ | [21] |
| 22 | $[Z, -Y, -X]$ | $\left[x+\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}-x\right]$ | [22] |
| 23 | $[-Z, Y, -X]$ | $\left[\frac{1}{2}-x, x+\frac{1}{2}, \frac{1}{2}-x\right]$ | [23] |
| 24 | $[-Z, -Y, X]$ | $\left[\frac{1}{2}-x, \frac{1}{2}-x, x+\frac{1}{2}\right]$ | [24] |

* Wyckoff site: 48f, site symmetry: 2..

Table 17: Wyckoff bond: 48a@48f, bond symmetry: 2..

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, X, Y]$ | $[x, 0, 0]$ | [1,-4] |
| 2 | $[0, -X, Y]$ | $[-x, 0, 0]$ | [2,-3] |
| 3 | $[Y, 0, X]$ | $[0, x, 0]$ | [5,-8] |
| 4 | $[Y, 0, -X]$ | $[0, -x, 0]$ | [6,-7] |
| 5 | $[X, Y, 0]$ | $[0, 0, x]$ | [9,-12] |
| 6 | $[-X, Y, 0]$ | $[0, 0, -x]$ | [10,-11] |
| 7 | $[X, 0, Y]$ | $\left[\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$ | [13,-16] |
| 8 | $[-X, 0, Y]$ | $\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$ | [14,-15] |
| 9 | $[0, Y, X]$ | $\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | [17,-20] |
| 10 | $[0, Y, -X]$ | $\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$ | [18,-19] |
| 11 | $[Y, X, 0]$ | $\left[\frac{1}{2}, \frac{1}{2}, x+\frac{1}{2}\right]$ | [21,-24] |
| 12 | $[Y, -X, 0]$ | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-x\right]$ | [22,-23] |

Table 18: Wyckoff bond: 48b@48f, bond symmetry: 2..

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, 0, 0]$ | $[x, 0, 0]$ | [1,4] |
| 2 | $[-X, 0, 0]$ | $[-x, 0, 0]$ | [2,3] |
| 3 | $[0, X, 0]$ | $[0, x, 0]$ | [5,8] |

*continued ...*

Table 18

| No. | vector | center | mapping |
|---|---|---|---|
| 4 | $[0, -X, 0]$ | $[0, -x, 0]$ | `[6,7]` |
| 5 | $[0, 0, X]$ | $[0, 0, x]$ | `[9,12]` |
| 6 | $[0, 0, -X]$ | $[0, 0, -x]$ | `[10,11]` |
| 7 | $[0, X, 0]$ | $\left[\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$ | `[13,16]` |
| 8 | $[0, -X, 0]$ | $\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$ | `[14,15]` |
| 9 | $[X, 0, 0]$ | $\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | `[17,20]` |
| 10 | $[-X, 0, 0]$ | $\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$ | `[18,19]` |
| 11 | $[0, 0, X]$ | $\left[\frac{1}{2}, \frac{1}{2}, x+\frac{1}{2}\right]$ | `[21,24]` |
| 12 | $[0, 0, -X]$ | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-x\right]$ | `[22,23]` |

Table 19: Wyckoff bond: `96c@48f`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $[x, 0, 0]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $[-x, 0, 0]$ | `[2]` |
| 3 | $[-X, Y, -Z]$ | $[-x, 0, 0]$ | `[3]` |
| 4 | $[X, -Y, -Z]$ | $[x, 0, 0]$ | `[4]` |
| 5 | $[Z, X, Y]$ | $[0, x, 0]$ | `[5]` |
| 6 | $[Z, -X, -Y]$ | $[0, -x, 0]$ | `[6]` |
| 7 | $[-Z, -X, Y]$ | $[0, -x, 0]$ | `[7]` |
| 8 | $[-Z, X, -Y]$ | $[0, x, 0]$ | `[8]` |
| 9 | $[Y, Z, X]$ | $[0, 0, x]$ | `[9]` |
| 10 | $[-Y, Z, -X]$ | $[0, 0, -x]$ | `[10]` |
| 11 | $[Y, -Z, -X]$ | $[0, 0, -x]$ | `[11]` |
| 12 | $[-Y, -Z, X]$ | $[0, 0, x]$ | `[12]` |
| 13 | $[Y, X, Z]$ | $\left[\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$ | `[13]` |
| 14 | $[-Y, -X, Z]$ | $\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$ | `[14]` |
| 15 | $[Y, -X, -Z]$ | $\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$ | `[15]` |
| 16 | $[-Y, X, -Z]$ | $\left[\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$ | `[16]` |
| 17 | $[X, Z, Y]$ | $\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | `[17]` |
| 18 | $[-X, Z, -Y]$ | $\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$ | `[18]` |
| 19 | $[-X, -Z, Y]$ | $\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$ | `[19]` |
| 20 | $[X, -Z, -Y]$ | $\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | `[20]` |
| 21 | $[Z, Y, X]$ | $\left[\frac{1}{2}, \frac{1}{2}, x+\frac{1}{2}\right]$ | `[21]` |
| 22 | $[Z, -Y, -X]$ | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-x\right]$ | `[22]` |
| 23 | $[-Z, Y, -X]$ | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-x\right]$ | `[23]` |
| 24 | $[-Z, -Y, X]$ | $\left[\frac{1}{2}, \frac{1}{2}, x+\frac{1}{2}\right]$ | `[24]` |

* Wyckoff site: `48g`, site symmetry: `2..`

Table 20: Wyckoff bond: `48a@48g`, bond symmetry: `2..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, X, Y]$ | $\left[x, \frac{1}{4}, \frac{1}{4}\right]$ | `[1,-4]` |
| 2 | $[0, -X, Y]$ | $\left[-x, \frac{3}{4}, \frac{1}{4}\right]$ | `[2,-3]` |

*continued ...*

Table 20

| No. | vector | center | mapping |
|---|---|---|---|
| 3 | $[Y, 0, X]$ | $\left[\frac{1}{4}, x, \frac{1}{4}\right]$ | `[5,-8]` |
| 4 | $[Y, 0, -X]$ | $\left[\frac{1}{4}, -x, \frac{3}{4}\right]$ | `[6,-7]` |
| 5 | $[X, Y, 0]$ | $\left[\frac{1}{4}, \frac{1}{4}, x\right]$ | `[9,-12]` |
| 6 | $[-X, Y, 0]$ | $\left[\frac{3}{4}, \frac{1}{4}, -x\right]$ | `[10,-11]` |
| 7 | $[X, 0, Y]$ | $\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$ | `[13,-16]` |
| 8 | $[-X, 0, Y]$ | $\left[\frac{1}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$ | `[14,-15]` |
| 9 | $[0, Y, X]$ | $\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$ | `[17,-20]` |
| 10 | $[0, Y, -X]$ | $\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{4}\right]$ | `[18,-19]` |
| 11 | $[Y, X, 0]$ | $\left[\frac{3}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$ | `[21,-24]` |
| 12 | $[Y, -X, 0]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$ | `[22,-23]` |

Table 21: Wyckoff bond: `48b@48g`, bond symmetry: `2..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, 0, 0]$ | $\left[x, \frac{1}{4}, \frac{1}{4}\right]$ | `[1,4]` |
| 2 | $[-X, 0, 0]$ | $\left[-x, \frac{3}{4}, \frac{1}{4}\right]$ | `[2,3]` |
| 3 | $[0, X, 0]$ | $\left[\frac{1}{4}, x, \frac{1}{4}\right]$ | `[5,8]` |
| 4 | $[0, -X, 0]$ | $\left[\frac{1}{4}, -x, \frac{3}{4}\right]$ | `[6,7]` |
| 5 | $[0, 0, X]$ | $\left[\frac{1}{4}, \frac{1}{4}, x\right]$ | `[9,12]` |
| 6 | $[0, 0, -X]$ | $\left[\frac{3}{4}, \frac{1}{4}, -x\right]$ | `[10,11]` |
| 7 | $[0, X, 0]$ | $\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$ | `[13,16]` |
| 8 | $[0, -X, 0]$ | $\left[\frac{1}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$ | `[14,15]` |
| 9 | $[X, 0, 0]$ | $\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$ | `[17,20]` |
| 10 | $[-X, 0, 0]$ | $\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{4}\right]$ | `[18,19]` |
| 11 | $[0, 0, X]$ | $\left[\frac{3}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$ | `[21,24]` |
| 12 | $[0, 0, -X]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$ | `[22,23]` |

Table 22: Wyckoff bond: `96c@48g`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $\left[x, \frac{1}{4}, \frac{1}{4}\right]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $\left[-x, \frac{3}{4}, \frac{1}{4}\right]$ | `[2]` |
| 3 | $[-X, Y, -Z]$ | $\left[-x, \frac{3}{4}, \frac{1}{4}\right]$ | `[3]` |
| 4 | $[X, -Y, -Z]$ | $\left[x, \frac{1}{4}, \frac{1}{4}\right]$ | `[4]` |
| 5 | $[Z, X, Y]$ | $\left[\frac{1}{4}, x, \frac{1}{4}\right]$ | `[5]` |
| 6 | $[Z, -X, -Y]$ | $\left[\frac{1}{4}, -x, \frac{3}{4}\right]$ | `[6]` |
| 7 | $[-Z, -X, Y]$ | $\left[\frac{1}{4}, -x, \frac{3}{4}\right]$ | `[7]` |
| 8 | $[-Z, X, -Y]$ | $\left[\frac{1}{4}, x, \frac{1}{4}\right]$ | `[8]` |
| 9 | $[Y, Z, X]$ | $\left[\frac{1}{4}, \frac{1}{4}, x\right]$ | `[9]` |
| 10 | $[-Y, Z, -X]$ | $\left[\frac{3}{4}, \frac{1}{4}, -x\right]$ | `[10]` |
| 11 | $[Y, -Z, -X]$ | $\left[\frac{3}{4}, \frac{1}{4}, -x\right]$ | `[11]` |
| 12 | $[-Y, -Z, X]$ | $\left[\frac{1}{4}, \frac{1}{4}, x\right]$ | `[12]` |
| 13 | $[Y, X, Z]$ | $\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$ | `[13]` |
| 14 | $[-Y, -X, Z]$ | $\left[\frac{1}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$ | `[14]` |

*continued ...*

Table 22

| No. | vector | center | mapping |
|---|---|---|---|
| 15 | $[Y, -X, -Z]$ | $\left[\frac{1}{4}, \frac{1}{2} - x, \frac{3}{4}\right]$ | [15] |
| 16 | $[-Y, X, -Z]$ | $\left[\frac{3}{4}, x + \frac{1}{2}, \frac{3}{4}\right]$ | [16] |
| 17 | $[X, Z, Y]$ | $\left[x + \frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$ | [17] |
| 18 | $[-X, Z, -Y]$ | $\left[\frac{1}{2} - x, \frac{3}{4}, \frac{1}{4}\right]$ | [18] |
| 19 | $[-X, -Z, Y]$ | $\left[\frac{1}{2} - x, \frac{3}{4}, \frac{1}{4}\right]$ | [19] |
| 20 | $[X, -Z, -Y]$ | $\left[x + \frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$ | [20] |
| 21 | $[Z, Y, X]$ | $\left[\frac{3}{4}, \frac{3}{4}, x + \frac{1}{2}\right]$ | [21] |
| 22 | $[Z, -Y, -X]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2} - x\right]$ | [22] |
| 23 | $[-Z, Y, -X]$ | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2} - x\right]$ | [23] |
| 24 | $[-Z, -Y, X]$ | $\left[\frac{3}{4}, \frac{3}{4}, x + \frac{1}{2}\right]$ | [24] |

* Wyckoff site: 96h, site symmetry: 1

Table 23: Wyckoff bond: 96a@96h, bond symmetry: 1

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $[x, y, z]$ | [1] |
| 2 | $[-X, -Y, Z]$ | $[-x, -y, z]$ | [2] |
| 3 | $[-X, Y, -Z]$ | $[-x, y, -z]$ | [3] |
| 4 | $[X, -Y, -Z]$ | $[x, -y, -z]$ | [4] |
| 5 | $[Z, X, Y]$ | $[z, x, y]$ | [5] |
| 6 | $[Z, -X, -Y]$ | $[z, -x, -y]$ | [6] |
| 7 | $[-Z, -X, Y]$ | $[-z, -x, y]$ | [7] |
| 8 | $[-Z, X, -Y]$ | $[-z, x, -y]$ | [8] |
| 9 | $[Y, Z, X]$ | $[y, z, x]$ | [9] |
| 10 | $[-Y, Z, -X]$ | $[-y, z, -x]$ | [10] |
| 11 | $[Y, -Z, -X]$ | $[y, -z, -x]$ | [11] |
| 12 | $[-Y, -Z, X]$ | $[-y, -z, x]$ | [12] |
| 13 | $[Y, X, Z]$ | $\left[y + \frac{1}{2}, x + \frac{1}{2}, z + \frac{1}{2}\right]$ | [13] |
| 14 | $[-Y, -X, Z]$ | $\left[\frac{1}{2} - y, \frac{1}{2} - x, z + \frac{1}{2}\right]$ | [14] |
| 15 | $[Y, -X, -Z]$ | $\left[y + \frac{1}{2}, \frac{1}{2} - x, \frac{1}{2} - z\right]$ | [15] |
| 16 | $[-Y, X, -Z]$ | $\left[\frac{1}{2} - y, x + \frac{1}{2}, \frac{1}{2} - z\right]$ | [16] |
| 17 | $[X, Z, Y]$ | $\left[x + \frac{1}{2}, z + \frac{1}{2}, y + \frac{1}{2}\right]$ | [17] |
| 18 | $[-X, Z, -Y]$ | $\left[\frac{1}{2} - x, z + \frac{1}{2}, \frac{1}{2} - y\right]$ | [18] |
| 19 | $[-X, -Z, Y]$ | $\left[\frac{1}{2} - x, \frac{1}{2} - z, y + \frac{1}{2}\right]$ | [19] |
| 20 | $[X, -Z, -Y]$ | $\left[x + \frac{1}{2}, \frac{1}{2} - z, \frac{1}{2} - y\right]$ | [20] |
| 21 | $[Z, Y, X]$ | $\left[z + \frac{1}{2}, y + \frac{1}{2}, x + \frac{1}{2}\right]$ | [21] |
| 22 | $[Z, -Y, -X]$ | $\left[z + \frac{1}{2}, \frac{1}{2} - y, \frac{1}{2} - x\right]$ | [22] |
| 23 | $[-Z, Y, -X]$ | $\left[\frac{1}{2} - z, y + \frac{1}{2}, \frac{1}{2} - x\right]$ | [23] |
| 24 | $[-Z, -Y, X]$ | $\left[\frac{1}{2} - z, \frac{1}{2} - y, x + \frac{1}{2}\right]$ | [24] |